Deep Residual Learning for Image Segmentation

Abstract
Medical image segmentation models often perform well on data from the same distribution as their training set. However, their performance may degrade when deployed in a different clinical environment. Domain shift refers to a change in the data distribution between source and target domains. In this paper we study this effect on three public datasets.

1. Introduction
Consider a model trained on images acquired at Hospital A. Images from Hospital B may differ in scanner settings, patient populations, or acquisition protocols. These differences can change image appearance without changing the underlying segmentation task.

2. Method

We adopt a U-Net backbone with residual blocks and train with Dice loss. Test-time augmentation is applied to estimate predictive uncertainty. Let x denote the input image and y the segmentation mask. We minimize L = 1 - 2|P ? G| / (|P| + |G|) over the training set.

3. Experiments

To assess generalization, we evaluate the model on independent target domain data and compare its performance with a source domain baseline. A change in image appearance alone does not tell us how much the segmentation performance will decrease; this requires experimental evidence. Results are summarized in Table 1.